Date: Day:

Categorical OR Indicator Variable

Categorical variables represent qualitative data, such as types, groups or regions, rather than continous measurements.

Model with 2 categories:

let

$$y = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 z_{1i} + \varepsilon_i$$

$$z_i = \begin{cases} 0 & \text{if } y_i \text{ from Male} \\ 1 & \text{if } y_i \text{ from Female} \end{cases}$$

$$x = \begin{bmatrix} 1 & x_{11} & x_{21} & 0 \\ 1 & \vdots & \vdots & 0 \\ 1 & \vdots & \vdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & & & 0 \\ 1 & x_{1N} & x_{2N} & 1 \end{bmatrix}$$

$$b = (X^T X)^{-1} (X^T Y)$$

EXAMPLE:

Table 12.7: Data for Example 12.9

x, (pH)	y, (amount of suspended solids)	Polymer
6.5	292	1
6.9	329	1
7.8	352	1
8.4	378	1
8.8	392	1
9.2	410	1
6.7	198	2
6.9	227	2
7.5	277	2
7.9	297	2
8.7	364	2
9.2	375	2
6.5	167	3
7.0	225	3
7.2	247	3
7.6	268	3
8.7	288	3
9.2	342	3

Model with 3 categories

For K categories use K − 1 indicator variables.

Example:

3 Polymer Types

let

$$\cancel{y = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 z_{1i} + \beta_4 z}$$

$$\hat{y} = \beta_0 + \beta_1 x_{1i} + \beta_2 z_{1i} + \beta_3 z_{2i}$$

let

$z_1 = 0$, $z_2 = 1$ for Polymer 1

$z_1 = 1$, $z_2 = 0$ for Polymer 2

$z_1 = 0$, $z_2 = 0$ for Polymer 3

X Matrix

$$\begin{bmatrix} 1 & 6.5 & 0 & 1 \\ 1 & 6.9 & 0 & 1 \\ 1 & 7.8 & 0 & 1 \\ 1 & 8.4 & 0 & 1 \\ 1 & 8.8 & 0 & 1 \\ 1 & 9.2 & 0 & 1 \\ 1 & 6.7 & 1 & 0 \\ 1 & 6.9 & 1 & 0 \\ 1 & 7.5 & 1 & 0 \\ 1 & 7.9 & 1 & 0 \\ 1 & 8.7 & 1 & 0 \\ 1 & 9.2 & 1 & 0 \\ 1 & 6.5 & 0 & 0 \\ 1 & 7.0 & 0 & 0 \\ 1 & 7.2 & 0 & 0 \\ 1 & 7.6 & 0 & 0 \\ 1 & 8.7 & 0 & 0 \\ 1 & 9.2 & 0 & 0 \end{bmatrix}$$